Plug-and-Play Multilingual Few-shot Spoken Words Recognition

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01 Context & approach

The challenge

Adding words or languages to a fixed-vocabulary recognizer usually requires retraining. *PLiX* learns an acoustic embedding space where a few example recordings define new words at inference time, while the encoder stays fixed.

Our contribution

Train spoken-word embeddings on over 12 million clips in 20 languages.
Test unseen words, five unseen languages and language-identification episodes.
Measure separate small/base encoders on three edge devices.

1 Learn from episodes

Train an encoder on support/query tasks; training and test vocabularies remain disjoint.

2 Average support embeddings

For each new word, average its labeled support embeddings to form a prototype; keep the encoder frozen.

3 Compare each query

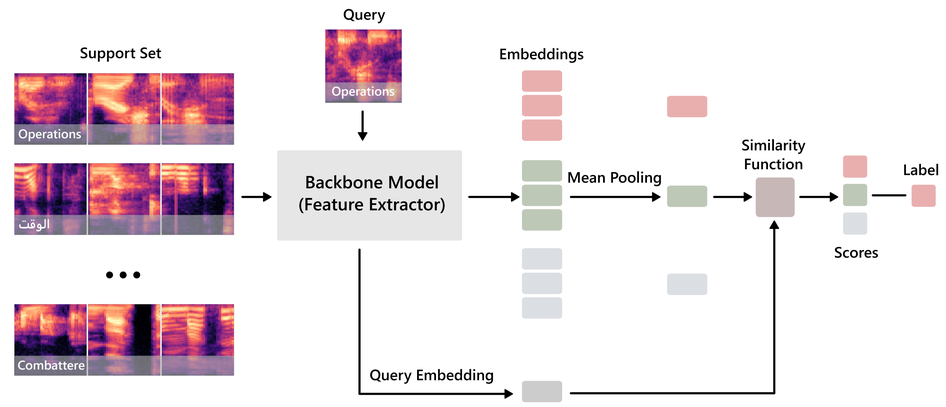
Encode the query directly and choose its nearest prototype by squared Euclidean distance; queries never enter support averaging.

Scope

Closed-set, segmented clips; continuous listening, unknown words and false triggers are not evaluated. Latency covers the encoder only; recognition accuracy after quantization is not reported.

02 Method

Recognize new spoken words from support examples without retraining the encoder.



Core formulation

$$\mathbf{p}_i = \frac{1}{|S_i|} \sum_{(x,y) \in S_i} f_\theta(x)$$

S_i contains support examples of word i; f_θ encodes each clip. Their mean p_i defines that word without updating θ.

$$\hat{y} = \arg\min_i \| f_\theta(x_q) - \mathbf{p}_i \|_2^2$$

Encode query x_q separately and choose the closest word prototype. Query embeddings are never averaged into p_i.

Paper **Code**

03 Results

95%
80-word recognition
English-only base; five supports/word

5.12 ms
small encoder latency
Quantized TinyNet-E; Pixel 7; encoder only

More support handles larger vocabularies

English test vocabulary	Supports per word	Accuracy (%) (Ours)
40 words	1	**90**
80 words	1	**86**
60 words	5	**96**
80 words	5	**95**

English-only base model on held-out MSWC words; 1,000 test episodes. Rounded mean accuracies.

Quantized encoder latency

Device	Small (ms)	Base (ms)
Pixel 7	5.12	78.07
Raspberry Pi 4	6.92	107.13
Jetson Nano	7.32	101.37

Quantized TinyNet-E / EfficientNet-v2 Medium; 200-run mean encoder latency. Excludes log-mel preprocessing and prototype classification.

Takeaway

New support examples add vocabulary without encoder retraining, with accuracy depending on language, vocabulary size and support count. The **5.12 ms** Pixel 7 result measures only the separate small encoder, excluding preprocessing and classification.